

Problem Set 71

Collection of Problems on Monotonicity and Extrema

May 1, 2026

Problem 1

Find the range of the following function using I/D test:

$$(b) \quad f(x) = \frac{1}{\pi}(\arcsin x + \arctan x) + \frac{x+1}{x^2+2x+5}$$

$$x \in [-1, 1]$$

$$f'(x) = \frac{1}{\pi} \left(\frac{1}{\sqrt{1-x^2}} + \frac{1}{1+x^2} \right) + \frac{-(x+3)(x-1)}{(x^2+2x+5)^2}$$

$$f'(x) > 0, x \in (-1, 1)$$

As $f'(x) > 0$ on $(-1, 1)$, $f(x)$ monotonically increases on this interval.

$$f(-1) = \frac{1}{\pi} \left(-\frac{\pi}{2} - \frac{\pi}{4} \right) + 0 = -\frac{3}{4}$$

$$f(1) = \frac{1}{\pi} \left(\frac{\pi}{2} + \frac{\pi}{4} \right) + \frac{2}{8} = 1$$

$$\boxed{f(x) \in \left[-\frac{3}{4}, 1\right]}$$

Problem 2

Find the number of roots of the function using I/D test: $f(x) = \frac{1}{(x+1)^3} - 3x + \sin x$.

$$f'(x) = -\frac{3}{(x+1)^4} - 3 + \cos x$$

$$f'(x) < 0, \quad x \in (-\infty, -1) \cup (-1, \infty)$$

$$f(-1^-) \rightarrow -\infty, f(-1^+) \rightarrow \infty$$

$$\boxed{2 \text{ roots}}$$

Problem 3

Use I/D test to prove that:

(a) $\ln(1+x) < x$ for $x > 0$

$$f(x) = x - \ln(1+x)$$

$$f(0) = 0 - \ln 1 = 0$$

$$f'(x) = 1 - \frac{1}{1+x} \implies f'(x) > 0$$

As $f'(x) > 0$ on $\mathbb{R}$, $f(x)$ monotonically increases on $\mathbb{R}$.

$$\boxed{x > \ln(1+x)}$$

(b) $\sin x < x$ and hence $\cos(\sin x) > \sin(\cos x)$ for $0 < x < \frac{\pi}{2}$

$$\begin{aligned} f(x) &= x - \sin x \\ f'(x) &= 1 - \cos x \\ x \in (0, \frac{\pi}{2}) &\implies f'(x) > 0 \end{aligned}$$

As $f'(x) > 0$ on $(0, \frac{\pi}{2})$, $f(x)$ monotonically increases on this interval.

$$\begin{aligned} &\boxed{x > \sin x} \\ &\cos x > \sin(\cos x) \\ &\sin(\frac{\pi}{2} - x) > \sin(\cos x) \\ &\sin(\frac{\pi}{2} - \sin x) > \sin(\frac{\pi}{2} - x) > \sin(\cos x) \\ &\boxed{\cos(\sin x) > \sin(\cos x)} \end{aligned}$$

Problem 4

(a) Find the relative maximum and/or minimum (if any) of the function $f(x) = (\frac{1}{x})^x$.

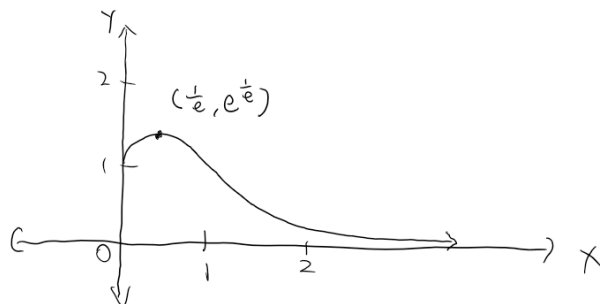
$$\begin{aligned} y &= (\frac{1}{x})^x \\ \ln y &= x \ln \frac{1}{x} \\ \frac{y'}{y} &= \ln \frac{1}{x} + x \cdot x \cdot -\frac{1}{x^2} \\ f'(x) &= (\frac{1}{x})^x (\ln \frac{1}{x} - 1) \\ f'(x) = 0 &\implies \ln \frac{1}{x} = 1 \\ &\boxed{x = \frac{1}{e}} \end{aligned}$$

As $f'(x)$ changes from positive to negative at $x = \frac{1}{e}$, $f(x)$ has a relative maximum at $x = \frac{1}{e}$.

(b) Sketch the graph of $f(x)$.

$$\lim_{x \rightarrow 0^+} f(x) = 1$$

$$\lim_{x \rightarrow \infty} f(x) = 0$$



(c) Find the range of $f(x)$.

$$x \in (0, \infty)$$

$$f\left(\frac{1}{e}\right) = e^{\frac{1}{e}}$$

$$\lim_{x \rightarrow \infty} f(x) = 0$$

$$\boxed{f(x) \in (0, e^{\frac{1}{e}}]}$$

(d) Determine which is bigger, $(\frac{1}{\pi})^{\frac{1}{e}}$ or $(\frac{1}{e})^{\frac{1}{\pi}}$?

$$\text{Define } f(x) = \left(\frac{1}{x}\right)^{\frac{1}{\pi+e-x}}$$

$$\ln f(x) = \frac{\ln \frac{1}{x}}{\pi + e - x}$$

$$\frac{f'(x)}{f(x)} = \frac{-\frac{\pi+e-x}{x} + \ln \frac{1}{x}}{(\pi + e - x)^2}$$

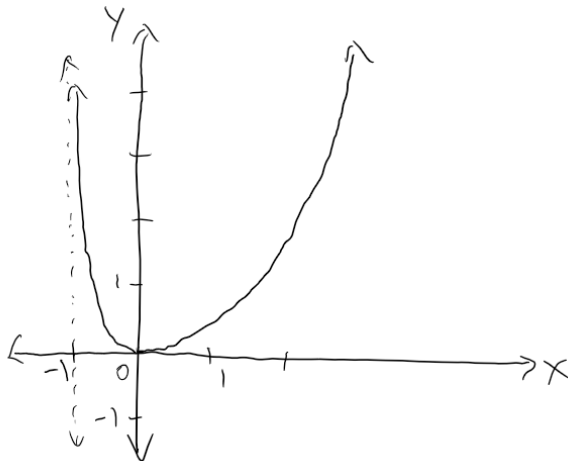
$$f'(x) = \left(\frac{1}{x}\right)^{\frac{1}{\pi+e-x}} \frac{-\frac{\pi+e-x}{x} - \ln x}{(\pi + e - x)^2}$$

$$x \in [e, \pi] \implies \ln x > 0, \frac{\pi + e - x}{x} > 0$$

$$f'(x) < 0, x \in (e, \pi)$$

As $f'(x) < 0$ on (e, π) , $f(x)$ monotonically decreases on this interval.

$$\boxed{\left(\frac{1}{e}\right)^{\frac{1}{\pi}} > \left(\frac{1}{\pi}\right)^{\frac{1}{e}}}$$

**Problem 5**

Sketch the graphs of the following functions using calculus :

(a) $y = \frac{x^2}{\sqrt{x+1}}$

$$D_f : (-1, \infty)$$

$$x\text{-intercept} : (0, 0)$$

$$y\text{-intercept} : (0, 0)$$

No Symmetry

$$\lim_{x \rightarrow -1^+} \frac{x^2}{\sqrt{x+1}} = \infty$$

$$f'(x) = \frac{2x\sqrt{x+1} - \frac{x^2}{2\sqrt{x+1}}}{x+1}$$

$$3x^2 + 4x = 0 \implies x = 0 \text{ or } -\frac{4}{3}$$

$$f'(x) < 0, \quad x \in (-1, 0)$$

$$f'(x) > 0, \quad x \in (0, \infty)$$

$$\min : f(0) = 0$$

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(b) $f(x) = \frac{x^2 - 5x + 6}{x^2 - x}$

$$D_f : \{x \in \mathbb{R} \mid x \neq 0, 1\}$$

$$y\text{-intercept} : (2, 0), (3, 0)$$

No Symmetry

$$\lim_{x \rightarrow -\infty} f(x) = 1$$

$$\lim_{x \rightarrow 0^-} f(x) = \infty$$

$$\lim_{x \rightarrow 0^+} f(x) = -\infty$$

$$\lim_{x \rightarrow 1^-} f(x) = -\infty$$

$$\lim_{x \rightarrow 1^+} f(x) = \infty$$

$$\lim_{x \rightarrow \infty} f(x) = 1$$

$$f'(x) = \frac{(2x-5)(x^2-x) - (2x-1)(x^2-5x+6)}{(x^2-x)^2}$$

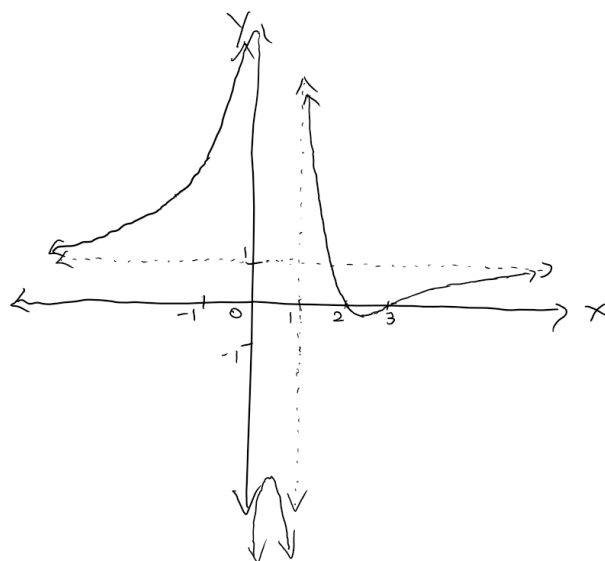
$$2x^2 - 6x + 3 = 0 \implies x = \frac{3 \pm \sqrt{3}}{2}$$

$$f'(x) < 0, \quad x \in \left(\frac{3-\sqrt{3}}{2}, \frac{3+\sqrt{3}}{2}\right)$$

$$f'(x) > 0, \quad x \in \left(-\infty, \frac{3-\sqrt{3}}{2}\right) \cup \left(\frac{3+\sqrt{3}}{2}, \infty\right)$$

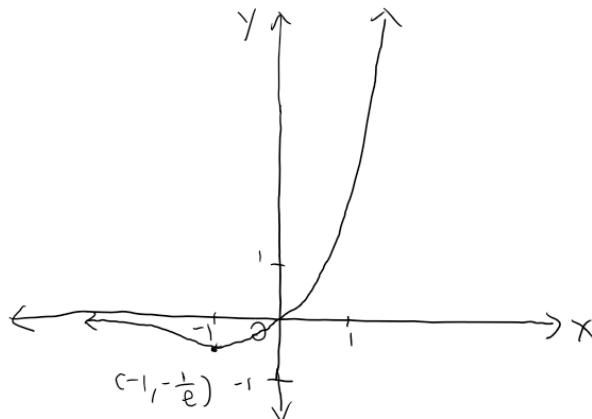
$$\text{min} : f\left(\frac{3+\sqrt{3}}{2}\right)$$

$$\text{max} : f\left(\frac{3-\sqrt{3}}{2}\right)$$



Problem 6

Sketch the graph of $f(x) = xe^x$ using calculus techniques. Then find:



(a) range of $f(x)$

$$f'(x) = e^x(x+1)$$

$$f'(x) = 0 \implies x = -1$$

As $f'(x)$ changes from negative to positive at $x = -1$, $f(x)$ has a local minimum at $x = -1$.

$$f(-1) = -\frac{1}{e}$$

$$\lim_{x \rightarrow -\infty} f(x) \rightarrow 0$$

$$\lim_{x \rightarrow \infty} f(x) \rightarrow \infty$$

$$f(x) \in \left[-\frac{1}{e}, \infty\right)$$

(b) point of inflection

$$f''(x) = e^x(x+2)$$

$$f''(x) = 0 \implies \boxed{x = -2}$$

(c) values of k such that $xe^x = k$ has two distinct real roots.

$$k \in \left(-\frac{1}{e}, 0\right)$$

Problem 7

Prove that there exist exactly two non-similar isosceles triangles ABC such that:

$$\tan A + \tan B + \tan C = 100$$

$$\begin{cases} A = B \\ A + B + C = \pi \end{cases}$$

$$\begin{aligned}
f(x) &= 2 \tan x + \tan(\pi - 2x) \\
&= 2 \tan x - \tan 2x \\
&= 2 \tan x - \frac{2 \tan x}{1 - \tan^2 x} \\
&= \frac{2 \tan^3 x}{\tan^2 x - 1}, \quad x \in (0, \frac{\pi}{2})
\end{aligned}$$

Let $\tan x = u$, $u \in (0, 1) \cup (1, \infty)$

$$\begin{aligned}
f(u) &= \frac{2u^3}{u^2 - 1} = 100 \\
u \in (0, 1) &\implies f(u) < 0 \\
&\begin{cases} \lim_{x \rightarrow 1^+} f(u) \rightarrow \infty \\ f(2) = \frac{16}{3} \\ \lim_{x \rightarrow \infty} f(u) \rightarrow \infty \end{cases}
\end{aligned}$$

By the intermediate value theorem, there exist exactly two values of $u \in (1, \infty)$ such that $f(u) = 100$. Hence, exactly two non-similar isosceles triangles ABC satisfies

$$\tan A + \tan B + \tan C = 100$$